

SPHERE: MISSION CONTROL

STUDENT MANUAL, EXPERIMENTAL METHOD AND ANALYSIS GUIDE

Experimental objective: Construct an insulated 100 mL water capsule, expose it to an ice-water boundary (0°C) and record the temperature for 15 minutes. Compare the measurements with Newton's law of cooling using residuals and MAE ($MAE = \frac{1}{n} \sum_{i=1}^n |\Delta T_i|$).

1. ENGINEERING CONTEXT & OBJECTIVES

Spacecraft thermal control must account for radiative exchange, internal heat generation and highly variable external heat fluxes. In vacuum there is no external convection; thermal design therefore differs fundamentally from this water-bath experiment.

Aerospace multilayer insulation uses low-emissivity reflective layers separated by low-conductance spacers. The cotton, bubble wrap and reflective film used here form an educational multilayer assembly, not flight-qualified MLI.

Test the assigned insulation configuration under a common boundary condition. Compare observations with the analytical reference curve and report residuals, MAE, uncertainty sources and model limitations.

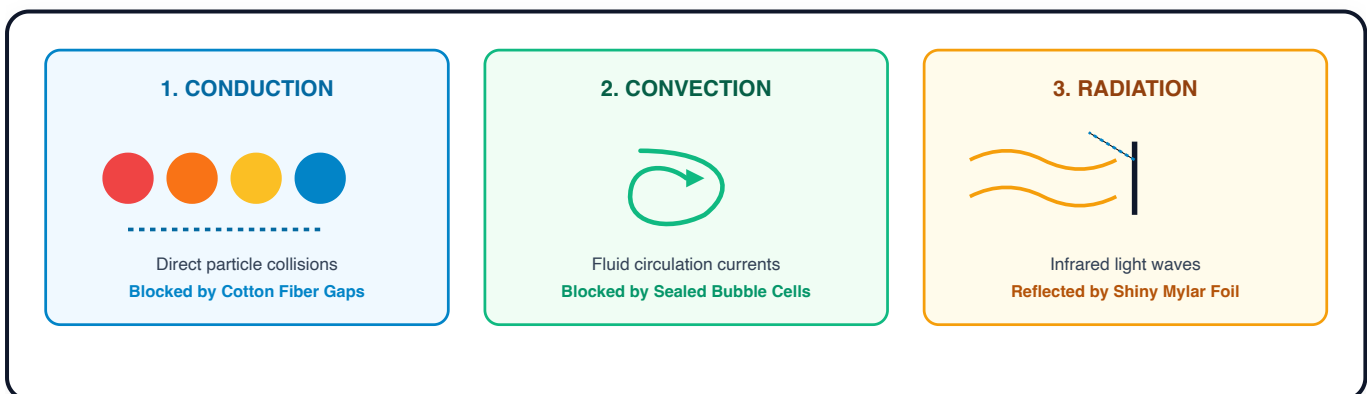


FIGURE 2: THREE MODES OF HEAT TRANSFER & THERMAL INSULATION BARRIERS

2. HARDWARE SETUP & SCHEMATIC

Assemble a sealed test capsule suitable for controlled immersion:

HARDWARE PART	DESCRIPTION	WHAT IT DOES IN OUR LAB
Core Jar	100 mL wide-mouth PET jar	Holds our hot water core ($T_0 = 80.0^{\circ}\text{C}$), representing the astronaut's internal body heat.
Cable Gland	PG7 waterproof nylon gland	Provides a compression seal around the thermometer cable; inspect and leak-test before use.
Digital Sensor	Waterproof digital thermometer	Measures internal core water temperature (T_{obs} , accuracy $\pm 0.5^{\circ}\text{C}$). Record the accuracy for your instrument.
Ice Bath Chamber	Ice-water slurry (0.0°C)	Provides a repeatable low-temperature boundary sink ($T_{\text{env}} = 0.0^{\circ}\text{C}$) for comparative testing.

Quick Assembly Steps:

- Feed the Sensor:** Unscrew the dome cap on the PG7 gland and slide the metal temperature probe through the gland body.
- Mount on Cap:** Push the threaded gland through the hole in the jar cap, and tighten the locknut on the underside so it won't budge.
- Tighten the Seal:** Hand-tighten the outer dome nut until the rubber ring clamps snugly around the cable. This stops any water from leaking in.
- Center the Probe:** Make sure the metal sensor tip hangs right in the center of the liquid without touching the bottom or sides.

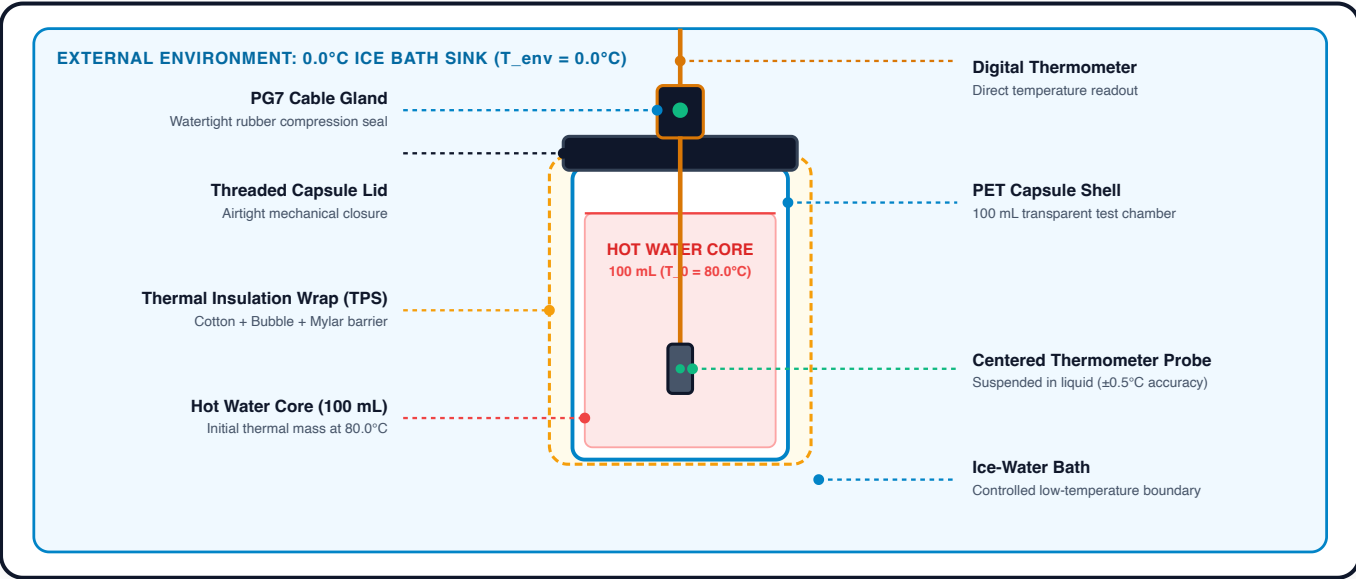


FIGURE 1: EXPERIMENTAL HARDWARE ASSEMBLY & SENSOR ALIGNMENT

3. THE 4 INSULATION CONFIGURATIONS

Each team tests one of four configurations under the same nominal boundary conditions ($T_0 = 80.0^\circ\text{C}$, $T_{\text{env}} = 0.0^\circ\text{C}$):



FIGURE 3: EDUCATIONAL MULTILAYER TEST ASSEMBLY

4. THE PHYSICS: NEWTON'S LAW OF COOLING

Heat always flows from warmer objects to cooler surroundings. Over 300 years ago, Sir Isaac Newton discovered that the rate at which an object cools is proportional to the temperature difference between the object and its environment:

NEWTON'S LAW OF COOLING (EXPONENTIAL DECAY MODEL)

$$T(t) = T_{\text{env}} + (T_0 - T_{\text{env}}) \cdot e^{-k \cdot t}$$

Where:

$T(t)$ = Predicted core temperature ($^{\circ}\text{C}$) at minute t

T_{env} = Ice bath temperature (0.0°C)

T_0 = Starting core temperature (80.0°C)

k = Cooling rate constant (min^{-1})

Interpretation of k : The fitted cooling constant k has units of min^{-1} . Under otherwise identical conditions, a smaller value corresponds to a slower approach to the bath temperature.

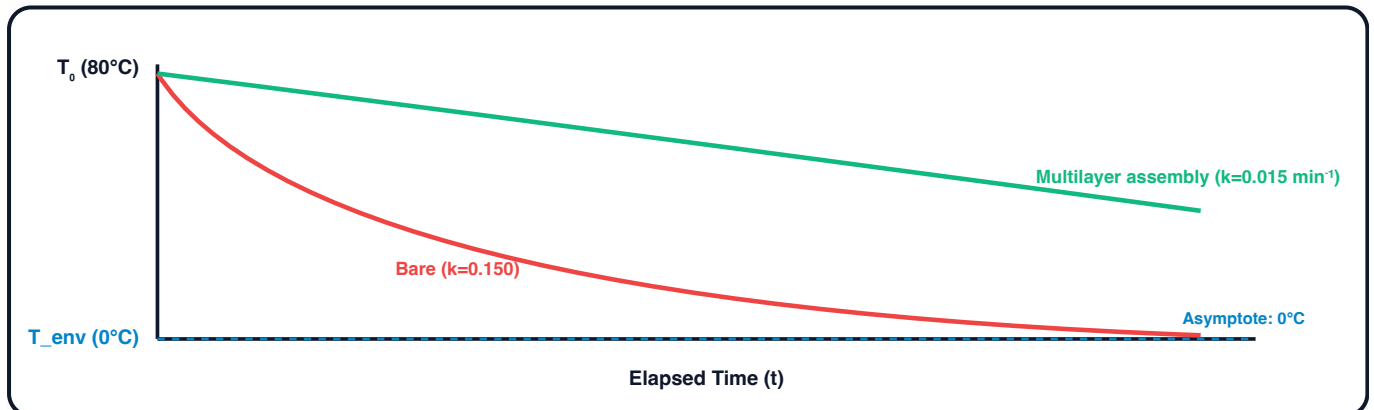


FIGURE 4: EXPONENTIAL DECAY PROFILES ACROSS HIGH VS LOW COOLING CONSTANTS (k)

5. STEP-BY-STEP LAB WORKFLOW

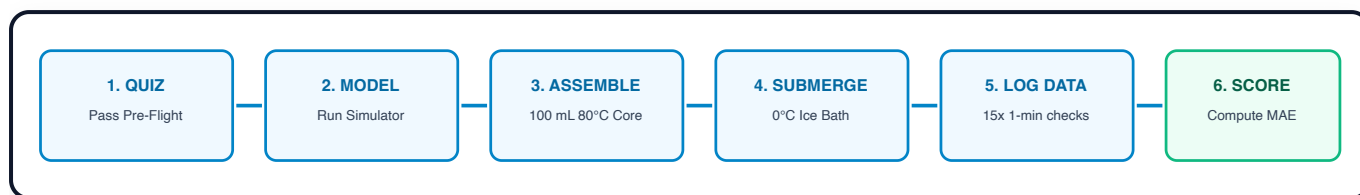


FIGURE 5: 6-STAGE EXPERIMENTAL PROTOCOL WORKFLOW

1. **Complete the concept check:** Review the heat-transfer questions before handling hot water.
2. **Generate Model Prediction:** Go to the **Simulator** tab, choose your team's assigned insulation configuration, verify starting conditions ($T_0 = 80.0^\circ\text{C}$, $T_{\text{env}} = 0.0^\circ\text{C}$), and click **RUN SIMULATION PREDICTION**.
3. **Fill the Water Core:** Carefully measure 100 mL of hot water at 80.0°C , pour it into your jar, screw the cap on tightly, and verify the seal is leak-free.
4. **Submerge and start timing:** Lower the capsule into the measured ice-water bath (0.0°C) and immediately start the timer in the **Measurement Log** tab.
5. **Log Temperature Every Minute:** Every 60 seconds for 15 minutes, write down the observed temperature in your Task Sheet and enter it into Mission Control.
6. **Evaluate:** Plot the measured and predicted curves, calculate residuals ($|\Delta T| = |T_{\text{obs}} - T_{\text{model}}|$) and MAE ($\text{MAE} = \frac{1}{n} \sum_{i=1}^n |\Delta T_i|$), and discuss the principal sources of uncertainty.

STUDENT NAME(S):

TEAM / TEST IDENTIFIER:

DATE / CLASS PERIOD:

TESTED INSULATION
CONFIGURATION:

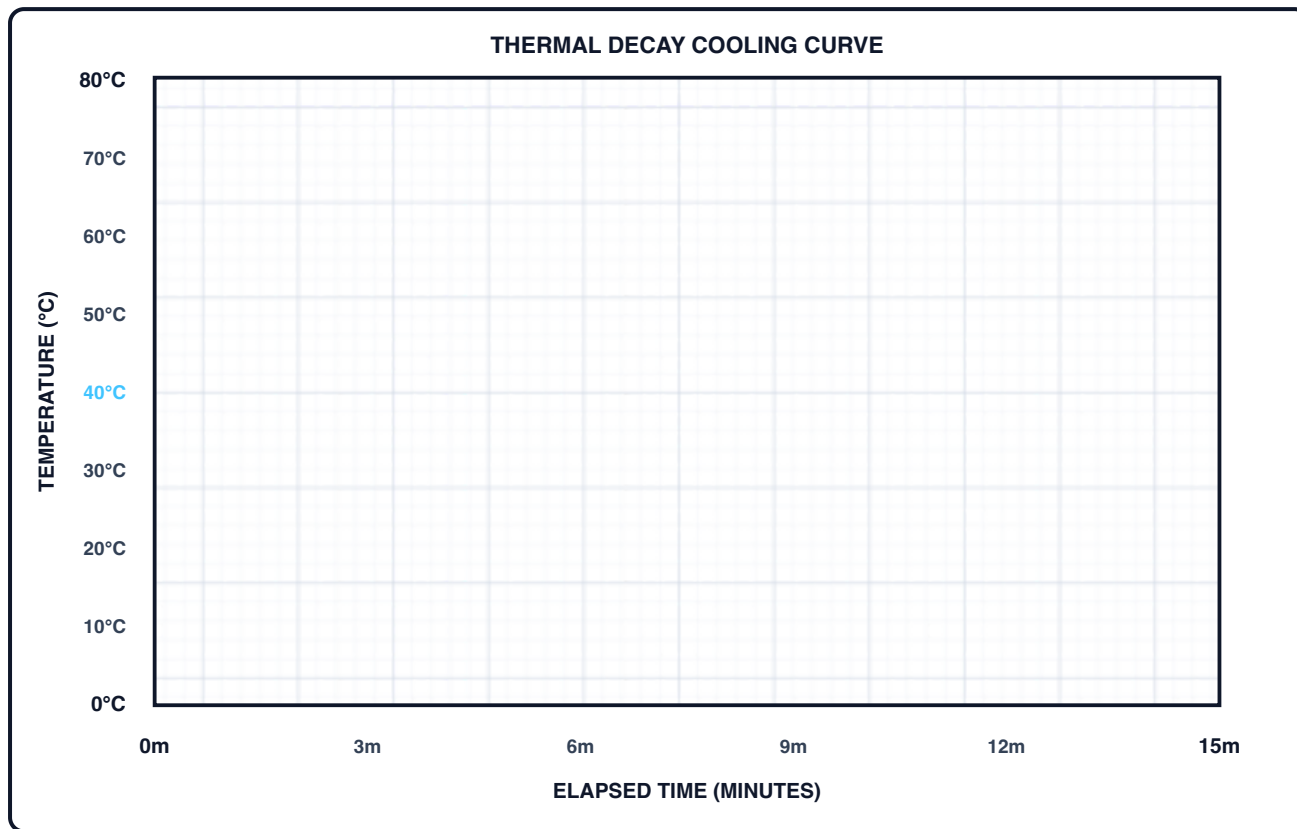
☐ A: Bare ☐ B: Cotton ☐
C: Reflective film ☐ D:
Multilayer

Part A. Experimental Temperature Log

TIME (min)	MEASURED TEMP (T_{obs} , °C)	PREDICTED TEMP (T_{model} , °C)	ABS DIFFERENCE $ \Delta T $ (°C)	OBSERVER NOTES
0 min	80.0°C	80.0°C	0.0°C	Capsule submerged in 0.0°C ice bath
1 min				Initial cooling rate check
2 min				
3 min				
4 min				
5 min				Mid-point check; check ice bath
6 min				
7 min				
8 min				
9 min				
10 min				Check curve flattening
11 min				
12 min				
13 min				
14 min				
15 min				Final measurement checkpoint

Part B. Coordinate Graph Grid

Plot your measured physical readings as solid dots (•) and connect them with a smooth line. Draw the theoretical model line as a dashed curve (---).



Part C. Residual and Mean Absolute Error Analysis

Calculation Steps:

- Sum of absolute residuals:** Add the n valid entries in the $|\Delta T|$ column: $\sum |\Delta T| = \sum_{i=1}^n |T_{\text{obs},i} - T_{\text{model},i}| = \underline{\hspace{2cm}}^{\circ}\text{C}$
- Mean absolute error:** Divide by the number of valid observations (n): $\text{MAE} = \frac{1}{n} \sum_{i=1}^n |T_{\text{obs},i} - T_{\text{model},i}| = \underline{\hspace{2cm}}^{\circ}\text{C}$
- Interpretation:** A lower MAE indicates closer agreement. Report the value with units ($^{\circ}\text{C}$) and discuss whether it is meaningful relative to thermometer accuracy ($\pm 0.5^{\circ}\text{C}$) and repeatability.

REPORT: MAE = $\underline{\hspace{2cm}}^{\circ}\text{C}$ | NUMBER OF OBSERVATIONS $n = \underline{\hspace{2cm}}$

7. LAB ANALYSIS & REFLECTION QUESTIONS

Question 1: Why doesn't shiny Mylar foil work by itself in cold water?

Mylar is famous for reflecting heat radiation in space. Why did Mylar alone perform poorly when dipped directly into the ice water bath (0.0°C) without cotton or bubble wrap underneath?

Question 2: What does the cooling constant (k) tell us?

In Newton's Law $T(t) = T_{\text{env}} + (T_0 - T_{\text{env}}) \cdot e^{-kt}$, does a *smaller* k value mean a better insulator or a worse insulator? Explain using the formula.

Question 3: How does Multi-Layer Insulation (MLI) work together?

Explain how combining Cotton + Bubble Wrap + Mylar in Configuration D ($k \approx 0.015 \text{ min}^{-1}$) creates a team effect that works far better than any single layer alone.

Question 4: Where could experimental errors come from?

List two physical reasons why your real sensor readings (T_{obs}) might differ from the smooth curve (T_{model}) predicted by the computer model.

Question 5: What happens if your ice bath warms up?

If your classroom ice bath warms from 0.0°C to 4.0°C while testing, what should you change in the Mission Control simulator so the prediction (T_{env}) stays accurate?

Question 6: What if we double the water volume?

If we doubled the water inside the jar from 100 mL to 200 mL, would it cool down faster or slower? How does thermal energy ($Q = m \cdot c \cdot \Delta T$) relate to this, and would k increase or decrease?

Question 7: Spacesuit Design Trade-Offs

If adding insulation layers keeps astronauts warmer, why don't space engineers just wrap a spacesuit in 50 layers of MLI? What problems would that cause for an astronaut on a spacewalk?

Question 8: How can measurement agreement be improved?

Describe two practical changes that could reduce MAE in a repeated experiment.